

Dr. Simona Cristea

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Head of AI and Genomics Data Science | Hale Center for Pancreatic Cancer
Faculty and Group Leader | Dana-Farber Cancer Institute
Research Scientist | Harvard School of Public Health

AI cancer evolution researcher developing and applying LLMs and transformer foundation models to genomics, with 10+ years experience in computational biology and healthcare. Led development of evoCancerGPT for cancer progression modeling and LLM systems analyzing 1M+ clinical records. Evaluated and benchmarked AI models for bioinformatics, with fine-tuning and interpretability analyses of single-cell LLMs on clinical trial genomics data. Built the Cristea Lab from scratch to 10 people in 2 years, combining AI research with genomics data science. 40+ publications (Nature, Nature Biotechnology, Cancer Discovery); \$2M+ funding awarded; 2 clinical trials using AI-driven genomic analyses. Public AI and science communication on [X](#); blogging on [Substack](#).

Leadership

- Team Building** Built the [Cristea Lab](#) from the ground-up to 10 people in 2 years, integrating AI scientists and computational biologists for cancer evolution modeling with genomics and data science expertise.
- Technical Leadership** Managing **\$1M+ annual budget** and computational infrastructure for AI/ML and genomics data science research across 20+ researchers. Established partnerships with Harvard, MIT, and clinical teams from Dana-Farber Cancer Institute and Massachusetts General Hospital to develop LLM tools analyzing 1M+ patient records and AI analyses for cancer clinical trial data.
- Research Impact** **40+ publications** (Nature, Nature Biotechnology, Cancer Discovery). Regular reviewer for top-tier journals. 30+ keynote presentations translating AI methods for life sciences, healthcare and genomics applications, including NeurIPS 2025.

AI Research

- AI for Cancer Evolution** Developed **evoCancerGPT**, a generative transformer for cancer progression modeling from single-cell genomics data. Built reasoning LLM systems (**DeepSeek-R1**) that outperform statistical models for biological data analysis (ICML 2025 GenAI in Bio).
- Clinical AI Systems** Implementing LLM-driven cancer risk stratification analyzing **1M+ longitudinal patient records** with integrated genomics data from Mass General-Brigham-Dana-Farber.
- Interpretability** Adapted genomics foundation models (**scGPT**) for cancer clinical trial genomics applications and provided mechanistic interpretations of attention, heads and layers.
- Biological LLMs** Benchmarked **reasoning LLMs** (DeepSeek-R1) against expert methodologies for zero-shot cell type annotation from single-cell genomics data.

Experience

- 2022 - now** **Head of Genomics Data Science and AI** at the Hale Center for Pancreatic Cancer Research, Dana-Farber Cancer Institute. Built from the ground up and leading the [Cristea Lab](#) (10 members) in 2 years, developing AI systems for cancer evolution modeling with genomics and clinical applications. Key work:
 - **Cancer evolution modeling**: Built evoCancerGPT transformer for single-cell genomics cancer progression; fine-tuned and interpreted scGPT for pancreatic cancer clinical trial analysis of genomics data
 - **AI in Bioinformatics**: Implemented reasoning LLMs (DeepSeek-R1) for biological and genomics data analysis from single-cell data
 - **Clinical AI**: Developed LLM systems analyzing 1M+ patient records for cancer risk prediction
 - **Spatial analysis**: Built analysis and AI platforms for spatial analysis of single cell transcriptomics and proteomics pancreatic cancer patient data
 - **Drug discovery**: CRISPR analysis pipelines for resistance mechanisms and target identification
 - **Immuno-genomics**: Multi-modal analysis of immunotherapy clinical trial data integrating temporal genomics and immune profiling

- 2022 - now **Research Scientist** (equivalent to Associate Professor) at Harvard School of Public Health
- 2020 - 2022 **Faculty and Group Leader** at Harvard Medical School. Built a research group (precursor to the Cristea Lab) focused on genomics cancer evolution and prevention:
- Developed genomics models for tumor evolution from multi-modal single-cell temporal data
 - Built spatio-temporal analysis frameworks for immunology cancer progression analysis
 - Built models integrating clinical with spatio-genomics trial data for triple-negative breast cancer Antibody Drug Conjugates (ADC) clinical trials
- 2016 - 2019 **Postdoc** at Dana-Farber and Harvard, developing methodology for single-cell genomics cancer evolution
- 2016 **Visiting postdoc** at Algorithmic Challenges in Genomics, Simmons Institute for the Theory of Computing at University of California, Berkeley and the Department of Biosystems, ETH Zürich
- 2011 - 2015 **PhD** in Computational Cancer Genomics at ETH Zürich, Department of Biosystems, developing computational models for cancer progression from genomics data

Education

- 2009 - 2011 **MSc in Computational Biology and Bioinformatics** at Department of Computer Science, ETH Zürich, Switzerland. Thesis: Monotone modules approximation
- 2008 - 2009 **Erasmus Exchange (BSc)** at University of Gothenburg, Sweden. Thesis: Economic policies for community dogs in Bucharest
- 2006 - 2009 **BSc in Mathematics** at Faculty of Mathematics & Computer Science, University of Bucharest, Romania. Thesis: Probabilistic models for transmission of genetic information
- 2006 - 2009 **BSc in Informatics** at Faculty of Cybernetics, Statistics and Informatics in Bucharest, Romania. Thesis: Optimizing stock portfolios via linear programming

Prizes, awards, fellowships (2M+ \$)

- 2025 - 2026 **Hale Cancer Center Award**: 1M \$ /year
- 2023 - 2024 **Hale Cancer Center Award**: 0.5M \$ /year
- 2022 Swiss National Science Foundation **Return Fellowship**: 122K \$ (renounced)
- 2021 - 2023 Dana-Farber Cancer Institute Faculty Support **Microgrant**: 20K \$
- 2019 Swiss National Science Foundation Postdoc Mobility **Fellowship**: 86K \$
- 2018 **Helen Gurley Brown Fellowship** at Dana-Farber Cancer Institute: 155K \$
- 2017 Swiss National Science Foundation Early Postdoc **Mobility Fellowship**: 100K \$
- 2010 **Best Information Processing Project** and **Gold Medal** at the textbfiGEM Competition for the project [E.lemming](#) in Cambridge, MA. Our team devised information processing algorithms to control the movement of genetically modified *E.coli*
- 2009 **ETH Zürich Excellence Scholarship**: top 5% Master-level students accepted by ETH Zürich: 36,5K USD
- 2009 **Full Merit Scholarship**, awarded by the Helsinki Summer School for the course Introduction to Bioinformatics, covering full tuition and boarding
- 2008 **Erasmus Scholarship**, awarded by The Faculty of Cybernetics, Statistics and Informatics and covering the full boarding and tuition costs for an Exchange semester abroad, at Gothenburg University, Sweden
- 2006 - 2009 **Excellence Scholarship**, awarded monthly without interruption for 3 years by The Faculty of Cybernetics, Statistics and Informatics to top 1% Bachelor students ranked by yearly scores
- 2006 **Merit Award** by the Romanian National Ministry of Education and Scientific Research to students who obtained perfect scores (10 / 10) in the National Baccalaureate (end of high-school exam)
- 2005 **Bronze Medal** at the Romanian National Mathematics Olympiad
- 2004 **Gold Medal** at the Romanian National Mathematics Olympiad, ranked 5th per country
- 2003 **Silver Medal** at the Romanian National Mathematics Olympiad

5 Recent Publications

- 2025 Xi Wang, Runzi Tan, Bo Wang, **Simona Cristea+**. Biological Reasoning with Reinforcement Learning through Natural Language Enables Generalizable Zero-Shot Cell Type Annotations. [bioRxiv](#) and accepted at **ICML 2025 GenBio Workshop**
- 2025 Haotian Cui, Alejandro Tejada-Lapuerta, Maria Brbic, Julio Saez-Rodriguez, **Simona Cristea**, Hani Goodarzi, Mohammad Lotfollahi, Fabian J. Theis, Bo Wang. [Multi-modal foundation models for molecular biology: opportunities and challenges](#). **Nature** 640, 623–633
- 2024 Runzi Tan, Haotian Cui, Bo Wang, Kimberly Perez, Andressa Dias Costa, Alexander Jordan, Thomas Karacic, Dalia Elganainy, Dan Y Gui, Suryun Kim, Chen Yuan, Morgan Truitt, Michael Downes, Ronald Evans, Tae Gyu Oh, Peter O'Dwyer, Andrew Aguirre, Jonathan A Nowak, Brian Wolpin, **Simona Cristea+**. [Evaluating and interpreting scGPT: A foundation model for single-cell biology in real-world cancer clinical trial data](#). **Abstract for AACR Special Conference on Advances in Pancreatic Cancer Research**. *Cancer Research* 84 (17): A029
- 2024 Triet Minh Bui, Ernesto A Rojas, Pierre Foidart, Zheqi Vaciry Li, Pengze Yan, Jun Nishida, Robert West, Shelley Hwang, **Simona Cristea**, Kornelia Polyak. [Cycling T regulatory cells architect an immune escape hub during in situ to invasive breast carcinoma transition](#). **Abstract for AACR Special Conference in Cancer Research**. *Cancer Immunology Research* 12(10): A028
- 2024 Julien Dilly, Megan T Hoffman, Laleh Abbassi, Ziyue Li, Francesca Paradiso, Brendan D Parent, Connor J Hennessey, Alexander C Jordan, Micaela Morgado, Shatavisha Dasgupta, Giselle A Uribe, Annan Yang, Kevin S Kapner, Felix P Hambitzer, Li Qiang, Hanrong Feng, Jacob Geisberg, Junning Wang, Kyle E Evans, Hengyu Lyu, Aislyn Schalck, Ningping Feng, Anastasia M Lopez, Christopher A Bristow, Michael P Kim, Kimal I Rajapakshe, Vahid Bahrambeigi, Jennifer A Roth, Kavita Garg, Paola A Guerrero, Ben Z Stanger, **Simona Cristea**, Scott W Lowe, Timour Baslan, Eliezer M Van Allen, Joseph D Mancias, Emily Chan, Abraham Anderson, Yuliya V Katlinskaya, Alex K Shalek, David S Hong, Shubham Pant, Jill Hallin, Kenna Anderes, Peter Olson, Timothy P Heffernan, Seema Chugh, James G Christensen, Anirban Maitra, Brian M Wolpin, Srivatsan Raghavan, Jonathan A Nowak, Peter S Winter, Stephanie K Dougan, Andrew J Aguirre. [Mechanisms of resistance to oncogenic KRAS inhibition in pancreatic cancer](#). **Cancer Discovery** 14(11)

All Publications

- 2025 Xi Wang, Runzi Tan, Bo Wang, **Simona Cristea+**. Biological Reasoning with Reinforcement Learning through Natural Language Enables Generalizable Zero-Shot Cell Type Annotations, **bioRxiv** BIORXIV/2025/659642 and accepted at **ICML GenBio 2025 Workshop**
- 2025 Haotian Cui, Alejandro Tejada-Lapuerta, Maria Brbic, Julio Saez-Rodriguez, **Simona Cristea**, Hani Goodarzi, Mohammad Lotfollahi, Fabian J. Theis, Bo Wang. [Multi-modal foundation models for molecular biology: opportunities and challenges](#). **Nature** 640, 623–633
- 2024 Runzi Tan, Haotian Cui, Bo Wang, Kimberly Perez, Andressa Dias Costa, Alexander Jordan, Thomas Karacic, Dalia Elganainy, Dan Y Gui, Suryun Kim, Chen Yuan, Morgan Truitt, Michael Downes, Ronald Evans, Tae Gyu Oh, Peter O'Dwyer, Andrew Aguirre, Jonathan A Nowak, Brian Wolpin, **Simona Cristea+**. [Evaluating and interpreting scGPT: A foundation model for single-cell biology in real-world cancer clinical trial data](#). **Abstract for AACR Special Conference on Advances in Pancreatic Cancer Research**. Cancer Research 84 (17): A029 (+ corresponding author)
- 2024 Julien Dilly, Megan T Hoffman, Laleh Abbassi, Ziyue Li, Francesca Paradiso, Brendan D Parent, Connor J Hennessey, Alexander C Jordan, Micaela Morgado, Shatavisha Dasgupta, Giselle A Uribe, Annan Yang, Kevin S Kapner, Felix P Hambitzer, Li Qiang, Hanrong Feng, Jacob Geisberg, Junning Wang, Kyle E Evans, Hengyu Lyu, Aislyn Schalck, Ningping Feng, Anastasia M Lopez, Christopher A Bristow, Michael P Kim, Kimal I Rajapakshe, Vahid Bahrampour, Jennifer A Roth, Kavita Garg, Paola A Guerrero, Ben Z Stanger, **Simona Cristea**, Scott W Lowe, Timour Baslan, Eliezer M Van Allen, Joseph D Mancias, Emily Chan, Abraham Anderson, Yuliya V Katlinskaya, Alex K Shalek, David S Hong, Shubham Pant, Jill Hallin, Kenna Anderes, Peter Olson, Timothy P Heffernan, Seema Chugh, James G Christensen, Anirban Maitra, Brian M Wolpin, Srivatsan Raghavan, Jonathan A Nowak, Peter S Winter, Stephanie K Dougan, Andrew J Aguirre. [Mechanisms of resistance to oncogenic KRAS inhibition in pancreatic cancer](#). **Cancer Discovery** 14(11)
- 2024 Laura M. Spring, Bogang Wu, Ting Liu, Jacob Geisberg, **Simona Cristea**, Veerle Bossuyt, Rachel Occhiogrosso Abelman, Nicole Peiris, James Coates, Siang-Boon Koh, Mengran Zhang, Lianne Ryan, Beverly Moy, Steven J. Isakoff, Sara M. Tolaney, Franziska Michor, Aditya Bardia, Leif W. Ellisen. [Intratumoral heterogeneity drives resistance to Antibody Drug Conjugate therapy: Analysis of the NeoSTAR trial of neoadjuvant Sacituzumab govitecan for localized TNBC](#). **Abstract for AACR Special Conference on Advances in Breast Cancer Research**. Cancer Research 84 (3): PR08
- 2024 Triet Minh Bui, Ernesto A Rojas, Pierre Foidart, Zheqi Vaciry Li, Pengze Yan, Jun Nishida, Robert West, Shelley Hwang, **Simona Cristea**, Kornelia Polyak. [Cycling T regulatory cells architect an immune escape hub during in situ to invasive breast carcinoma transition](#). **Abstract for AACR Special Conference in Cancer Research**. Cancer Immunology Research 12(10): A028
- 2024 Yang Liu, Jennifer Altreuter, Sudheshna Bodapati, **Simona Cristea**, Cheryl J Wong, Catherine J Wu, Franziska Michor. [Predicting patient outcomes after treatment with immune checkpoint blockade: A review of biomarkers derived from diverse data modalities](#). **Cell Genomics** 4
- 2023 Jun Nishida*, **Simona Cristea*+**, Sudheshna Bodapati*, Julieann Puleo, Gali Bai, Ashka Patel, Melissa Hughes, Craig Snow, Virginia Borges, Kathryn J Ruddy, Laura C Collins, Anne-Marie Feeney, Kara Slowik, Veerle Bossuyt, Deborah Dillon, Nancy U Lin, Ann H Partridge, Franziska Michor+, Kornelia Polyak+. [Peripheral blood TCR clonotype diversity as an age-associated marker of breast cancer progression](#). **PNAS** 120(49); *co-first; +co-senior author
- 2023 Zhihong Chen, Bruno Giotti, Milota Kaluzova, Cameron Herting, Gonzalo Pinero, **Simona Cristea**, James L Ross, James Ackley, Victor Maximov, Frank Szulzewsky, Mar Marquez-Roper, Angelo Angione, Noah Nichols, Nadejda Tsankova, Franziska Michor, Dmitry M Shayakhmetov, David H Gutmann, Alexander M Tsankov, Dolores Hambardzumyan. [A paracrine circuit of IL-1b and IL-1R1 between myeloid and tumor cells drives glioblastoma progression](#). **Journal of Clinical Investigation** 133:22
- 2023 Nicole Peiris, **Simona Cristea**, Mengran Zhang, Siang Koh, James Coates, Ilze Smidt, Veerle Bossuyt, Laura Spring, Aditya Bardia, Leif Ellisen. [Utilizing scRNA sequencing to understand biomarkers of response and resistance to Sacituzumab Govitecan in localized TNBC](#). **Abstract for AACR Annual Meeting**. Cancer Research 83: 4353

- 2023 Michael S. Hoetker, Masaki Yagi, Bruno Di Stefano, Justin Langerman, **Simona Cristea**, Lai Ping Wong, Aaron J. Huebner, Jocelyn Charlton, Weixian Deng, Chuck Haggerty, Ruslan I. Sadreyev, Alexander Meissner, Franziska Michor, Kathrin Plath, Konrad Hochedlinger. [H3K36 methylation maintains cell identity by regulating opposing lineage programs](#). **Nature Biotechnology** 25(8)
- 2023 Kacper A. Walentynowicz*, Dalit Engelhardt*, **Simona Cristea***, Shreya Yadav, Ugoma Onubogu, Roberto Salatino, Cristina Vincentelli, Thomas O. McDonald, Franziska Michor, Michalina Janiszewska. [Single-cell heterogeneity of EGFR and CDK4 co-amplification is linked to immune infiltration in glioblastoma](#). **Cell Reports** 42(3); *co-first author
- 2022 Masa Aleckovic*, **Simona Cristea***, Carlos R. Gil Del Alcazar*, Pengze Yan, Lina Ding, Ethan D. Krop, Nicholas W. Harper, Ernesto Rojas Jimenez, Donghao Lu, Anushree C. Gulvady, Pierre Foidart, Marco Seehawer, Benedetto Diciaccio, Katherine C. Murphy, Jason Pyrdol, Jayati Anand, Kodie Garza, Kai W. Wucherpennig, Rulla Tamimi, Franziska Michor, Kornelia Polyak. [Breast cancer prevention by short-term inhibition of TGF \$\beta\$ signaling](#). **Nature Communications** 13:7558; *co-first author
- 2022 Rick ACM Boonen, Wouter W Wiegant, Nandi Celosse, Bas Vroling, Stephan Heijl, Zsafia Kote-Jarai, Martina Mijuskovic, **Simona Cristea**, Nienke Solleveld-Westerink, Tom van Wezel, Niko Beerenwinkel, Rosalind Eeles, Peter Devilee, Maaïke PG Vreeswijk, Giancarlo Marra, Haico van Attikum. [Functional analysis identifies damaging CHEK2 missense variants associated with increased cancer risk](#). **Cancer Research** 82(4):615–631
- 2021 Siang-Boon Koh, Brian N Dontchos, Veerle Bossuyt, Christine Edmonds, **Simona Cristea**, Nsan Melkonjan, Lindsey Mortensen, Annie Ma, Cassidy Beyerlin, Elyssa Denault, Elizabeth Niehoff, Taghreed Hirz, David B Sykes, Franziska Michor, Michelle Specht, Constance Lehman, Leif W Ellisen, Laura M Spring. [Systematic tissue collection during clinical breast biopsy is feasible, safe and enables high-content translational analyses](#). **npj Precision Oncology** 5:85
- 2021 Masaki Yagi, Fei Ji, Jocelyn Charlton*, **Simona Cristea***, Kathleen Messemer, Naftali Horwitz, Bruno Di Stefano, Nikolaos Tsooulidis, Michael S Hoetker, Aaron J Huebner, Ori Bar-Nur, Albert E Almada, Masakazu Yamamoto, Anthony Patelunas, David J Goldhamer, Amy J Wagers, Franziska Michor, Alexander Meissner, Ruslan I Sadreyev, Konrad Hochedlinger. [Dissecting dual roles of MyoD during lineage conversion to mature myocytes and myogenic stem cells](#). **Genes & Development** 35:1209-1228; *equal contribution
- 2021 Michalina Janiszewska, Shayna Stein, Otto Metzger Filho, Jennifer Eng, Natalie L. Kingston, Nicholas W. Harper, Inga H. Rye, Maša Aleckovic, Anne Trinh, Katherine C. Murphy, Elisabetta Marangoni, **Simona Cristea**, Benjamin Oakes, Eric P. Winer, Ian E. Krop, Hege G. Russnes, Paul T. Spellman, Elmar Bucher, Zhi Hu, Koei Chin, Joe W. Gray, Franziska Michor, Kornelia Polyak. [The impact of tumor epithelial and microenvironmental heterogeneity on treatment responses in HER2+ breast cancer](#). **JCI insight** 16(11):e147617
- 2020 Patrick Grossmann, **Simona Cristea**, Niko Beerenwinkel. [Clonal evolution driven by superdriver mutations](#). **BMC Evolutionary Biology** 20(1),1-11
- 2019 Michalina Janiszewska, Doris P Tabassum, Zafira Castano, **Simona Cristea**, Kimiyo N Yamamoto, Natalie L Kingston, Katherine C Murphy, Shaokun Shu, Nicholas W Harper, Carlos Gil Del Alcazar, Masa Aleckovic, Muhammad B Ekram, Ofir Cohen, ..., Andriy Marusyk, Kai W Wucherpennig, Nikhil Wagle, Rong Fan, Franziska Michor, Sandra S McAllister, Kornelia Polyak. [Subclonal cooperation drives metastasis by modulating local and systemic immune microenvironments](#). **Nature Cell Biology** 21(7),879-888
- 2019 Dana Silverbush*+, **Simona Cristea*+**, Gali Yanovich, Tamar Geiger, Niko Beerenwinkel, Roded Sharan. [Simultaneous Integration of multi-omics data improves the identification of cancer driver modules](#). **Cell Systems** 8(5),456-466.e5; *co-first; +co-senior author
- 2019 Lina Ding, Ying Su, Anne Fassl, Kunihiro Hinohara, Xintao Qiu, Nicholas W Harper, Sung Jin Huh, Noga Bloushtain-Qimron, Bojana Jovanovic, Muhammad Ekram, Xiaoyuan Zi, William C Hines, Masa Aleckovic, Carlos Gil del Alcazar, Ryan J Caulfield, Dennis M Bonal, Quang-De Nguyen, Vanessa F Merino, Sibgat Choudhury, Gabrielle Ethington, Laura Panos, Michael Grant, William Herlihy, Alfred Au, Gedge D Rosson, Pedram Argani, Andrea L Richardson, Deborah Dillon, D Craig Allred, Kirsten Babski, Elizabeth Min Hui Kim, Charles H McDonnell, Jon Wagner, Ron Rowberry, Kristie Bobolis, Celina G Kleer, E Shelley Hwang, Joanne L Blum, **Simona Cristea**, Piotr Sicinski, Rong Fan, Henry W Long, Saraswati Sukumar, So Yeon Park, Judy E Garber, Mina Bissell, Jun Yao, Kornelia Polyak. [Perturbed myoepithelial cell differentiation in BRCA mutation carriers and in ductal carcinoma in situ](#). **Nature Communications** 10(1),1-16

- 2019 Aron Gyuris, Jose Navarrete-Perea, Ala Jo, **Simona Cristea**, Shuang Zhou, Kyle Fraser, Zhiyun Wei, Anna M Krichevsky, Ralph Weissleder, Hakho Lee, Steve P Gygi, Al Charest. [Physical and molecular landscapes of mouse glioma extracellular vesicles define heterogeneity](#). **Cell Reports** 27(13),3972-3987.e6
- 2018 Mihriban Karaayvaz*, **Simona Cristea***, Shawn M. Gillespie, Anoop P. Patel, Ravi Mylvaganam, Christina C. Luo, Michelle C. Specht, Bradley E. Bernstein, Franziska Michor, Leif W. Ellisen. [Unravelling subclonal heterogeneity and aggressive disease states in TNBC through single-cell RNA-seq](#). **Nature Communications** 9(1),3588; *co-first author
- 2018 **Simona Cristea**, Kornelia Polyak. [Dissecting the mammary gland one cell at a time](#). **Nature Communications** 9(1),2473
- 2018 Dana Silverbush*+, **Simona Cristea*+**, Gali Yanovich, Tamar Geiger, Niko Beerenwinkel, Roded Sharan. [Simultaneous integration of multi-omics data improves the identification of cancer driver modules](#). Research in Computational Molecular Biology Conference 2018 (**RECOMB**) in Paris, France; highlight paper at Intelligent Systems for Molecular Biology Conference 2018 (**ISMB**) in Chicago, IL (*co-first; +co-corresponding author)
- 2018 Hyunghoon Cho, Bonnie Berger, Jian Peng, Cyril Galitzine, Olga Vitek, Pierre M. Jean Beltran, Ileana M. Cristea, Franziska Gortler, Stefan Solbrig, Tilo Wettig, Peter J. Oefner, Rainer Spang, Michael Altenbuchinger, Rebecca Sarto Basso, Dorit Hochbaum, Fabio Vandin, Dana Silverbush, **Simona Cristea**, Gali Yanovich, Tamar Geiger, Niko Beerenwinkel, Roded Sharan, Zhemin Zhou, Nina Luhmann, Nabil-Fareed Alikhan, Mark Achtman [Principles of Systems Biology, No. 31](#). **Cell Systems**. Volume 7, Issue 2, P133-135
- 2017 Priyanka G Bhosale, Manishkumar Pandey, **Simona Cristea**, Mickey Shah, Asawari Patil, Niko Beerenwinkel, Alejandro A. Schäffer, Manoj B Mahimkar. [Recurring amplification at 11q22.1-q22.2 locus plays an important role in lymph node metastasis and radioresistance in OSCC](#). **Scientific Reports** 7 (1), 16051
- 2017 **Simona Cristea**, Jack Kuipers, Niko Beerenwinkel. [pathTiMEx: Joint inference of mutually exclusive cancer pathways and their progression dynamics](#). **Journal of Computational Biology** 24 (6), 603-615
- 2017 **Simona Cristea**, Jack Kuipers, Niko Beerenwinkel. [pathTiMEx: Joint inference of mutually exclusive cancer pathways and their progression dynamics](#). Research in Computational Molecular Biology Conference 2016 (**RECOMB**) in Santa Monica, CA
- 2017 Priyanka G. Bhosale*, **Simona Cristea***, Srikant Ambatipudi, Rajiv S. Desai, Rajiv Kumar, Asawari Patil, Shubhada Kane, Anita M. Borges, Alejandro A. Schäffer, Niko Beerenwinkel, Manoj B. Mahimkar. [Chromosomal alterations and gene expression changes associated with the progression of leukoplakia to advanced gingivobuccal cancer](#). **Translational Oncology** 10 (3), 396-409 (*co-first author)
- 2016 Melanie M. Erzinger, Cedric Bovet, Katrin M. Hecht, Sabine Senger, Pascale Winiker, Nadine Sobotzki, **Simona Cristea**, Niko Beerenwinkel, Jerry W. Shay, Giancarlo Marra, Bernd Wollscheid, Shana J. Sturla. [Sulforaphane preconditioning sensitizes human colon cancer cells towards the bio-reductive anticancer pro-drug PR-104A](#). **PLoS ONE** 11(3)
- 2016 Computational studies in cancer progression and treatment. Doctoral Thesis at ETH Zurich. **Library of ETH Zurich**
- 2016 **Simona Constantinescu**, Ewa Szczurek, Pejman Mohammadi, Jörg Rahnenführer, Niko Beerenwinkel. [TiMEx: A waiting time model for mutually exclusive cancer alterations](#). **Bioinformatics** 32(7)
- 2015 **Simona Constantinescu**, Ewa Szczurek, Pejman Mohammadi, Jörg Rahnenführer, Niko Beerenwinkel. [TiMEx: A waiting time model for mutually exclusive cancer alterations](#). Research in Computational Molecular Biology Satellite Workshop 2015 (**RECOMB Cancer Computational Biology**) in Warsaw, Poland
- 2015 Atilgan Yilmaz, Rachel Engeler, **Simona Constantinescu**, Konstantinos D Kokkaliaris, Christos Dimitrakopoulos, Timm Schroeder, Niko Beerenwinkel and Renato Paro. [Ectopic Expression of *Mx2* in mammalian myotubes recapitulates aspects of amphibian muscle dedifferentiation](#). **Stem Cell Research** 15(3)
- 2015 **Simona Constantinescu**, Shana J. Sturla, Giancarlo Marra, Bernd Wollscheid, Niko Beerenwinkel. [Computational data integration in toxicogenomics](#). In **Computational Systems Toxicology** 371-392, Methods in Pharmacology and Toxicology

- 2014 **Simona Constantinescu**, Katrin Hecht, Nadine Sobotzki, Melanie M Erzinger, Cedric Bovet, Jerry W Shay, Bernd Wollscheid, Shana J Sturla, Giancarlo Marra, Niko Beerenwinkel. [Transcriptomic responses of cancerous and noncancerous human colon cells to sulforaphane and selenium](#). **Chemical Research in Toxicology** 27 (3), 377-386
- 2011 Monotone module approximation. Master Thesis at ETH Zurich. **Library of ETH Zurich**
- 2011 Hans-Michael Kaltenbach, **Simona Constantinescu**, Justin Feigelman, Jörg Stelling. [Graph-based decomposition of biochemical reaction networks into monotone subsystems](#) In **International Workshop on Algorithms in Bioinformatics** 139-150, Lecture Notes in Computer Science

Submitted manuscripts

- 2025 Dalia Elganainy, Alexander C. Jordan, Andressa Dias Costa, Suryun Kim, Sung Chul Hong, Dan Y. Gui, Xi Wang, Chen Yuan, Xinran Qi, Hannah L. Williams, Anders Ohman, Jennifer S. Thalappillil, Sudheshna Bodapati, Sara A. Vayrynen, Vasilena Gocheva, Paul Oberstein, Diane M. Simeone, Peter J. O'Dwyer, Thomas B. Karasic, Douglas A. Robinson, Kimberly J. Perez, James M. Cleary, Brandon M. Huffman, Rishi Surana, Joseph D. Mancias, Harshabad Singh, Eliezer Van Allen, Richard F. Dunne, Thomas Clancy, David C. Linehan, Daniel T. Chang, Aram F. Hezel, Albert C. Koong, **Simona Cristea**, Andrew J. Aguirre, Brian M. Wolpin, Jonathan A. Nowak. *Spatially-resolved protein and RNA expression analyses define fibroblast organization, reveal functional specialization, and uncover chemotherapy-associated changes in pancreatic cancer*. Under review by Cancer Discovery
- 2025 Xi Wang, Runzi Tan, Bo Wang, **Simona Cristea**+. *Biological reasoning with reinforcement learning through natural language enables generalizable zero-shot cell type annotations*. Submitted to Nature. (+ corresponding author)

Contributed Code and Data

- 2025 **BioReasoning with Reinforcement Learning**, a benchmarking study showing how the reasoning LLM DeepSeek-R1 outperforms expert statistical models at scRNAseq cell typing: [GitHub](#), [bioRxiv](#) and accepted at [ICML 2025 GenBio Workshop](#).
- 2025 **evoCancerGPT**, a generative pre-trained transformer decoder-only single-cell foundation model designed to forecast future gene expression profiles in cancer evolution by leveraging previous cell states at the level of single patients: [GitHub](#), manuscript under submission.
- 2025 Single cell COSMX **spatial transcriptomics pipeline** for pancreatic cancer tumor analyses: [GitHub](#).
- 2024 **Hex Aligner**, a tool for spatial image analysis to align cancer images with different offsets, as well as tessellate images into hexagons for downstream analyses: [GitHub](#).
- 2024 **Hale Center Data Repository**, a data resource for downloading and exploring pancreatic cancer samples treated with immunotherapy at DFCI: [Webapp](#) and [GitHub](#). The webapp is only accessible within the Dana-Farber network.
- 2023 Single-cell heterogeneity of EGFR and CDK4 co-amplification is linked to immune infiltration in glioblastoma: [GitHub](#), [dbGaP](#), [paper](#) and [zenodo](#).
- 2022 Breast cancer prevention by short-term inhibition of TGFB signaling: [GitHub](#), [GEO](#), [paper](#) and [zenodo](#).
- 2021 Systematic tissue collection during clinical breast biopsy is feasible, safe and enables high-content translational analyses: [GEO](#) and [paper](#).
- 2021 Dissecting dual roles of MyoD during lineage conversion to mature myocytes and myogenic stem cells: [GEO](#) and [paper](#).
- 2019 Subclonal cooperation drives metastasis by modulating local and systemic immune microenvironments: [GEO](#) and [paper](#).
- 2018 Unravelling subclonal heterogeneity and aggressive disease states in TNBC through single-cell RNA-seq: [GitHub](#), [GEO](#) and [paper](#).
- 2018 **ModulOmics**, a multi-modal probabilistic graphical model for cancer module identification, integrating multiple genomics data sources: [GitHub](#) and [paper](#).
- 2016 **pathTiMEx**, a waiting time probabilistic graphical model and computational framework for the simultaneous inference of cancer pathways and their progression dynamics from cancer genomics data: [GitHub](#) and [paper](#).
- 2015 **TiMEx**, a waiting time probabilistic graphical model and the computational framework for inferring cancer pathways from cancer genomics data, following an evolutionary waiting time model: [GitHub](#) and [paper](#).
- 2015 Ectopic expression of Msx2 in mammalian myotubes recapitulates aspects of amphibian muscle dedifferentiation: [GEO](#) and [paper](#).